



Seat No.	
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Set	P
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**S.E. (Civil) (Part – I) (New – CBCS Pattern) Examination, 2018
ENGINEERING GEOLOGY**

Day and Date : Monday, 17-12-2018
Time : 10.00 a.m. to 1.00 p.m.

Total Marks : 70

- Instructions :**
- 1) Figures to the **right** indicate **full** marks.
 - 2) Draw neat labelled diagrams **wherever** necessary.
 - 3) Q. No. **1** is **compulsory**. It should be solved in **first 30 minutes** in Answer Book Page No. **3**. **Each** question carries **one** mark.
 - 4) **Answer MCQ/Objective type questions on Page No. 3 only. Don't forget to mention, Q.P. Set (P/Q/R/S) on Top of Page.**

MCQ/Objective Type Questions

Duration : 30 Minutes

Marks : 14

1. Choose the correct answer :

14

- 1) Metamorphic rock with alternating layers of light and dark minerals is called as a _____
a) Schist b) Gneiss c) Marble d) Slate
- 2) Which one of the following is not a part of volcano ?
a) dyke b) crater c) conduit d) delta
- 3) The solid products of volcano are called _____
a) lava b) magma c) pyroclasts d) phenocrysts
- 4) Hardness of corundum is _____
a) 10 b) 9 c) 8 d) 7
- 5) A coarse grained sedimentary rock with rounded pebbles, cobbles is called as _____
a) conglomerate b) breccias c) sandstone d) shale

P.T.O.



- 6) Sloping surface of the valley upon which dam rests is known as _____
a) abutment b) toe c) heel d) gallery
- 7) The primary force driving landslides is _____
a) tidal b) solar c) gravitational d) geo-thermal
- 8) The capacity of the rocks to withstand bending loads is called as _____
a) tensile strength b) compressive strength
c) durability d) bulk density
- 9) Bolting, slope treatment and construction of retaining wall are prevention methods of _____
a) landslides b) earthquake c) tsunamis d) volcano
- 10) The length of the core obtained is called _____
a) core loss b) core run c) core recovery d) none of these
- 11) Tsunamis are generated by activity of _____ below the ocean.
a) volcano b) earthquake c) tunneling d) all of these
- 12) The color of building stone mainly depends upon _____
a) texture b) structure
c) strength d) mineral composition
- 13) The most efficient, cheaper and rapid method of determination of sub-surface geology is _____
a) seismic method b) gravity method
c) magnetic method d) electric resistivity method
- 14) Cross bedding is commonly found in _____
a) limestone b) shale c) sandstone d) basalt
- _____



Seat No.	
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**S.E. (Civil) (Part – I) (New – CBCS Pattern) Examination, 2018
ENGINEERING GEOLOGY**

Day and Date : Monday, 17-12-2018

Marks : 56

Time :10.00 a.m. to 1.00 p.m.

- Instructions :** 1) Q. 6 and Q. 11 are **compulsory**.
2) Figures to the right indicate **full** marks.
3) Draw neat-labelled diagrams **wherever** necessary.

SECTION – I

2. A) Define fold. Describe symmetrical and asymmetrical folds. **6**

B) Define igneous rocks. Describe their any two structures. **6**

OR

3. A) Describe Horst and Graben fault. **6**

B) Describe physical properties and chemical properties of quartz group minerals. **6**

4. Define landslide. Describe different causes of land sliding. **7**

OR

5. Define earthquake. Describe geological effects of earthquake. **7**

6. Write notes on (**any three**) : **9**

- a) Residual sedimentary rocks
- b) Acidic and basic lava
- c) Dynamothermal metamorphism
- d) Types of discontinuity within the earth
- e) Fault mountain.

Set P



SECTION – II

7. A) Define Dams. Describe desirable and undesirable conditions for construction of dams on inclined beds. **6**
- B) What are building stones ? Add a note on building stones of India. **6**
- OR
8. A) Explain various types of dams. **6**
- B) Define tunnels. Explain tunneling through horizontal beds. **6**
9. What are bridges ? Explain various types of bridges. **7**
- OR
10. Describe in detail geological aspects of tunneling through folds. **7**
11. Write notes on (**any three**) : **9**
- a) Crushing strength of building stones
 - b) Silting process
 - c) Electric resistivity method
 - d) RQD
 - e) Core loss and core recovery.
-

Set **Q**

Day and Date : Monday, 17-12-2018
Time :10.00 a.m. to 1.00 p.m.

Total Marks : 70

Instructions :

- 1) Figures to the **right** indicate **full** marks.
- 2) Draw neat labelled diagrams **wherever** necessary.
- 3) Q. No. **1** is **compulsory**. It should be solved in **first 30 minutes** in Answer Book Page No. **3**. **Each** question carries **one** mark.
- 4) **Answer MCQ/Objective type questions on Page No. 3 only. Don't forget to mention, Q.P. Set (P/Q/R/S) on Top of Page.**

MCQ/Objective Type Questions

Duration : 30 Minutes

Marks : 14

1. Choose the correct answer :

14

- 1) The capacity of the rocks to withstand bending loads is called as _____.
 - a) tensile strength
 - b) compressive strength
 - c) durability
 - d) bulk density
- 2) Bolting, slope treatment and construction of retaining wall are prevention methods of _____.
 - a) landslides
 - b) earthquake
 - c) tsunamis
 - d) volcano
- 3) The length of the core obtained is called _____.
 - a) core loss
 - b) core run
 - c) core recovery
 - d) none of these
- 4) Tsunamis are generated by activity of _____ below the ocean.
 - a) volcano
 - b) earthquake
 - c) tunneling
 - d) all of these

P.T.O.



- 5) The color of building stone mainly depends upon _____
a) texture b) structure
c) strength d) mineral composition
- 6) The most efficient, cheaper and rapid method of determination of sub-surface geology is _____
a) seismic method b) gravity method
c) magnetic method d) electric resistivity method
- 7) Cross bedding is commonly found in _____
a) limestone b) shale c) sandstone d) basalt
- 8) Metamorphic rock with alternating layers of light and dark minerals is called as a _____
a) Schist b) Gneiss c) Marble d) Slate
- 9) Which one of the following is not a part of volcano ?
a) dyke b) crater c) conduit d) delta
- 10) The solid products of volcano are called _____
a) lava b) magma c) pyroclasts d) phenocrysts
- 11) Hardness of corundum is _____
a) 10 b) 9 c) 8 d) 7
- 12) A coarse grained sedimentary rock with rounded pebbles, cobbles is called as _____
a) conglomerate b) breccias c) sandstone d) shale
- 13) Sloping surface of the valley upon which dam rests is known as _____
a) abutment b) toe c) heel d) gallery
- 14) The primary force driving landslides is _____
a) tidal b) solar c) gravitational d) geo-thermal



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- Instructions :** 1) Q. 6 and Q. 11 are **compulsory**.
2) Figures to the right indicate **full** marks.
3) Draw neat-labelled diagrams **wherever** necessary.

SECTION – I

2. A) Define fold. Describe symmetrical and asymmetrical folds. **6**

B) Define igneous rocks. Describe their any two structures. **6**

OR

3. A) Describe Horst and Graben fault. **6**

B) Describe physical properties and chemical properties of quartz group minerals. **6**

4. Define landslide. Describe different causes of land sliding. **7**

OR

5. Define earthquake. Describe geological effects of earthquake. **7**

6. Write notes on (**any three**) : **9**

- a) Residual sedimentary rocks
- b) Acidic and basic lava
- c) Dynamothermal metamorphism
- d) Types of discontinuity within the earth
- e) Fault mountain.

Set Q



SECTION – II

7. A) Define Dams. Describe desirable and undesirable conditions for construction of dams on inclined beds. **6**
B) What are building stones ? Add a note on building stones of India. **6**
OR
8. A) Explain various types of dams. **6**
B) Define tunnels. Explain tunneling through horizontal beds. **6**
9. What are bridges ? Explain various types of bridges. **7**
OR
10. Describe in detail geological aspects of tunneling through folds. **7**
11. Write notes on (**any three**) : **9**
a) Crushing strength of building stones
b) Silting process
c) Electric resistivity method
d) RQD
e) Core loss and core recovery.
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Seat No.	
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Set	R
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**S.E. (Civil) (Part – I) (New – CBCS Pattern) Examination, 2018
ENGINEERING GEOLOGY**

Day and Date : Monday, 17-12-2018
Time : 10.00 a.m. to 1.00 p.m.

Total Marks : 70

- Instructions :**
- 1) Figures to the **right** indicate **full** marks.
 - 2) Draw neat labelled diagrams **wherever** necessary.
 - 3) Q. No. **1** is **compulsory**. It should be solved in **first 30 minutes** in Answer Book Page No. **3**. **Each** question carries **one** mark.
 - 4) **Answer MCQ/Objective type questions on Page No. 3 only. Don't forget to mention, Q.P. Set (P/Q/R/S) on Top of Page.**

MCQ/Objective Type Questions

Duration : 30 Minutes

Marks : 14

1. Choose the correct answer :

14

- 1) A coarse grained sedimentary rock with rounded pebbles, cobbles is called as _____
a) conglomerate b) breccias c) sandstone d) shale
- 2) Sloping surface of the valley upon which dam rests is known as _____
a) abutment b) toe c) heel d) gallery
- 3) The primary force driving landslides is _____
a) tidal b) solar c) gravitational d) geo-thermal
- 4) The capacity of the rocks to withstand bending loads is called as _____
a) tensile strength b) compressive strength
c) durability d) bulk density
- 5) Bolting, slope treatment and construction of retaining wall are prevention methods of _____
a) landslides b) earthquake c) tsunamis d) volcano

P.T.O.



- 6) The length of the core obtained is called _____.
a) core loss b) core run c) core recovery d) none of these
- 7) Tsunamis are generated by activity of _____ below the ocean.
a) volcano b) earthquake c) tunneling d) all of these
- 8) The color of building stone mainly depends upon _____.
a) texture b) structure
c) strength d) mineral composition
- 9) The most efficient, cheaper and rapid method of determination of sub-surface geology is _____.
a) seismic method b) gravity method
c) magnetic method d) electric resistivity method
- 10) Cross bedding is commonly found in _____.
a) limestone b) shale c) sandstone d) basalt
- 11) Metamorphic rock with alternating layers of light and dark minerals is called as a _____.
a) Schist b) Gneiss c) Marble d) Slate
- 12) Which one of the following is not a part of volcano ?
a) dyke b) crater c) conduit d) delta
- 13) The solid products of volcano are called _____.
a) lava b) magma c) pyroclasts d) phenocrysts
- 14) Hardness of corundum is _____.
a) 10 b) 9 c) 8 d) 7



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S.E. (Civil) (Part – I) (New – CBCS Pattern) Examination, 2018
ENGINEERING GEOLOGY

Day and Date : Monday, 17-12-2018
Time :10.00 a.m. to 1.00 p.m.

Marks : 56

- Instructions :** 1) Q. 6 and Q. 11 are **compulsory**.
2) Figures to the right indicate **full** marks.
3) Draw neat-labelled diagrams **wherever** necessary.

SECTION – I

2. A) Define fold. Describe symmetrical and asymmetrical folds. **6**

B) Define igneous rocks. Describe their any two structures. **6**

OR

3. A) Describe Horst and Graben fault. **6**

B) Describe physical properties and chemical properties of quartz group minerals. **6**

4. Define landslide. Describe different causes of land sliding. **7**

OR

5. Define earthquake. Describe geological effects of earthquake. **7**

6. Write notes on (**any three**) : **9**

- a) Residual sedimentary rocks
- b) Acidic and basic lava
- c) Dynamothermal metamorphism
- d) Types of discontinuity within the earth
- e) Fault mountain.

Set R



SECTION – II

7. A) Define Dams. Describe desirable and undesirable conditions for construction of dams on inclined beds. **6**
B) What are building stones ? Add a note on building stones of India. **6**
OR
8. A) Explain various types of dams. **6**
B) Define tunnels. Explain tunneling through horizontal beds. **6**
9. What are bridges ? Explain various types of bridges. **7**
OR
10. Describe in detail geological aspects of tunneling through folds. **7**
11. Write notes on (**any three**) : **9**
a) Crushing strength of building stones
b) Silting process
c) Electric resistivity method
d) RQD
e) Core loss and core recovery.
-

Set **S**

Day and Date : Monday, 17-12-2018
Time : 10.00 a.m. to 1.00 p.m.

Total Marks : 70

Instructions :

- 1) Figures to the **right** indicate **full** marks.
- 2) Draw neat labelled diagrams **wherever** necessary.
- 3) Q. No. **1** is **compulsory**. It should be solved in **first 30 minutes** in Answer Book Page No. **3**. **Each** question carries **one** mark.
- 4) **Answer MCQ/Objective type questions on Page No. 3 only. Don't forget to mention, Q.P. Set (P/Q/R/S) on Top of Page.**

MCQ/Objective Type Questions

Duration : 30 Minutes

Marks : 14

1. Choose the correct answer :

14

- The length of the core obtained is called _____.
a) core loss b) core run c) core recovery d) none of these
- Tsunamis are generated by activity of _____ below the ocean.
a) volcano b) earthquake c) tunneling d) all of these
- The color of building stone mainly depends upon _____.
a) texture b) structure
c) strength d) mineral composition
- The most efficient, cheaper and rapid method of determination of sub-surface geology is _____.
a) seismic method b) gravity method
c) magnetic method d) electric resistivity method
- Cross bedding is commonly found in _____.
a) limestone b) shale c) sandstone d) basalt

P.T.O.



- 6) Metamorphic rock with alternating layers of light and dark minerals is called as a _____
a) Schist b) Gneiss c) Marble d) Slate
- 7) Which one of the following is not a part of volcano ?
a) dyke b) crater c) conduit d) delta
- 8) The solid products of volcano are called _____
a) lava b) magma c) pyroclasts d) phenocrysts
- 9) Hardness of corundum is _____
a) 10 b) 9 c) 8 d) 7
- 10) A coarse grained sedimentary rock with rounded pebbles, cobbles is called as _____
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- 11) Sloping surface of the valley upon which dam rests is known as _____
a) abutment b) toe c) heel d) gallery
- 12) The primary force driving landslides is _____
a) tidal b) solar c) gravitational d) geo-thermal
- 13) The capacity of the rocks to withstand bending loads is called as _____
a) tensile strength b) compressive strength
c) durability d) bulk density
- 14) Bolting, slope treatment and construction of retaining wall are prevention methods of _____
a) landslides b) earthquake c) tsunamis d) volcano
- _____



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SECTION – I

2. A) Define fold. Describe symmetrical and asymmetrical folds. **6**

B) Define igneous rocks. Describe their any two structures. **6**

OR

3. A) Describe Horst and Graben fault. **6**

B) Describe physical properties and chemical properties of quartz group minerals. **6**

4. Define landslide. Describe different causes of land sliding. **7**

OR

5. Define earthquake. Describe geological effects of earthquake. **7**

6. Write notes on (**any three**) : **9**

- a) Residual sedimentary rocks
- b) Acidic and basic lava
- c) Dynamothermal metamorphism
- d) Types of discontinuity within the earth
- e) Fault mountain.

Set S



SECTION – II

7. A) Define Dams. Describe desirable and undesirable conditions for construction of dams on inclined beds. **6**
B) What are building stones ? Add a note on building stones of India. **6**
OR
8. A) Explain various types of dams. **6**
B) Define tunnels. Explain tunneling through horizontal beds. **6**
9. What are bridges ? Explain various types of bridges. **7**
OR
10. Describe in detail geological aspects of tunneling through folds. **7**
11. Write notes on (**any three**) : **9**
a) Crushing strength of building stones
b) Silting process
c) Electric resistivity method
d) RQD
e) Core loss and core recovery.
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